

Xiaoqing PAN

xiaoqing.pan@uni-jena.de | [LinkedIn](#) | [ORCID](#) | [Website](#) | +49 15563309561 | Jena, Germany

Re: postdoc position

Dear Prof. Dr. Huarte,

My name is Xiaoqing, I just finished my PhD study at Hans-Knöll-Institute in Jena, Germany, expect to have my defense in summer 2026. My PhD studied how pathogenic microorganisms produce, process, and deploy RNAs to regulate its biology. Through this work, I developed a strong interest in how RNA shapes the homeostasis of the cells.

During my PhD, I investigated RNA regulation on both sides of host-fungal interactions. I participated in a project investigating transcriptomic responses in human PBMCs, which provided me with experience in immune cell biology and strengthened my interest in inflammatory signaling pathways. Through characterizing transcriptomic responses after fungal stimulations in human PBMCs with a focus on alternative splicing and ADAR1-dependent A-to-I RNA editing patterns, we surprisingly found out that one fungus can induce RNA editing as well as disrupt host RNA splicing. Further, when we investigated the disrupted RNA, we found out that host pathogen recognition receptor NLRC5 is truncated after fungal stimulation. Work has gained me experiences analyzing multiple RNA-seq datasets, including DEG analysis, alternative splicing analysis, RNA structure analysis and RNA editing analysis.

In parallel, I was also interested in fungal small RNAs, dsRNA metabolism. In *Aspergillus fumigatus*, a human fungal pathogen, I analyzed sRNA-seq and mRNA-seq data to characterize the small RNA landscape and the contribution of RNAi machinery to fungal biology. After the bioinformatic prediction, we further use qPCR as well as Northern blot for the experimental validation. We surprisingly found out that too much double-stranded RNA may arrest the fungal growth, which is widely studied in cancer research. This work gave me practical experience with RNA data analysis, transcriptomics, target-oriented thinking, and the connection between RNA regulation and microbial behavior. Work has resulted in two publications on *RNA* and *Molecular microbiology*.

In addition to my work on fungal pathogens, I am attracted to your research because it combines RNA biology, cell biology, and functional genomics to address fundamental questions in cancer. I am willing by the possibility of contributing my expertise in sequencing-based approaches, molecular biology while expanding my skills in RNA-FISH, scRNA-seq and many others to ovarian cancer research. This will make me a more comprehensive scientist, more than a fungal expertise.

Thank you for your time and consideration.

Best,

Xiaoqing

Xiaoqing PAN

xiaoqing.pan@uni-jena.de | [LinkedIn](#) | [ORCID](#) | [Website](#) | +49 15563309561 | Jena, Germany

SUMMARY

RNA biologist and computational transcriptomics researcher with expertise in high-throughput sequencing, gene regulation, and RNA-mediated cellular responses. Experienced in integrating molecular biology with bioinformatics to study small RNAs, RNA interference, alternative splicing, and RNA editing across host and microbial systems. Proficient in RNA-seq analysis, reproducible computational workflows (R, Bash), and sequencing-based approaches for investigating post-transcriptional regulation.

TECHNICAL SKILLS

- **Molecular Biology:** sRNA cloning, NGS library preparation, GUS reporter assays, luciferase and GFP reporter assays, Gibson Assembly Cloning, Genetic Engineering, Northern Blot, qPCR, PCR, immunostaining, LC-MS, cultivation of microorganisms (BSL-1/2), plant cultivation and tissue preparation
- **Bioinformatics:** RNA-seq, sRNA-seq, differential expression, alternative splicing, small RNA characterization and target prediction, data visualization, HPC (Slurm), UCSC, SwissProt, KEGG, NCBI
- **Programming:** R, Python, Bash, Git, Docker, Linux
- **Language:** Chinese (native), English (fluent), German (B1)

EXPERIENCES

RNA landscape of human fungal pathogen *Aspergillus fumigatus*

- Characterized RNAi and tRNA fragments processing pathways in *A. fumigatus* across developmental stages (conidia, mycelium), revealing morphotype-specific RNA regulatory mechanisms
- Analyzed differential expressed genes on mRNA-, sRNA-seq datasets, uncovering post-transcriptional regulatory networks controlling fungal development and stress response.
- Validated genes of interest (sRNA, tRNA fragments, sRNA targets) via qPCR, northern blot

Human PBMC transcriptome changes upon fungal stimulations

- Analyzed pathogen-specific transcriptomic signatures in human immune cells responding to fungal pathogens, characterized differential expressed genes, alternative splicing, A-to-I RNA editing, pathogen recognition receptor pathway activation
- Built HPC pipeline, wrote custom R scripts for multi-condition cohort analysis

Human neutrophils exporting extracellular RNA upon *A. fumigatus* confrontation

- Analyzed EV-associated sRNA, tRNA, and mRNA cargo from primary human neutrophils during fungal challenge

Non-coding RNA analysis of rice-*Xanthomonas oryzae* interaction

- Studied how T3SS-dependent bacterial effectors from *Xanthomonas oryzae* shape host non-coding RNA regulation; performed GUS reporter assays, plant cultivation, tissue preparation, LC-MS, qPCR, and PCR
- Engineered overexpressed and knockout mutants of circular RNA and microRNA in rice, established genetic tools that enabled investigation of ncRNA-mediated regulation

Teaching

- Microbiology and Molecular Biology, MMB005, 2022-2025. Supervised master students investigating gene regulation together with Dr. Blango and other colleagues.

EDUCATION

PhD | Microbiology, Friedrich Schiller University, Germany (supervisor: Dr. Matthew Blango) 2026

Thesis: RNA regulation in host-fungal interactions: investigation of the host epitranscriptome and fungal small RNAs

MSc | Bioengineering, Chinese Academy of Sciences, China (supervisor: Dr. Kuaifei Xia) 2021

Thesis: Non-coding RNAs ciR133 and miR168a-5p play roles in response to Xoo and abiotic stress in rice, respectively

PUBLICATIONS

- **Pan X***, Kelani AA*, Schrettenbrunner L, Prabakar S, Israni B, Blango MG. (2026). An improved description of the small RNA landscape of the human fungal pathogen *Aspergillus fumigatus*. Mol Microbiol 1-12.
- **Pan X***, Bruch A*, Blango MG. (2024). The past, present, and future of RNA modifications in infectious disease research. ACS Infect Dis 10(12), 4017-4029.
- Xia K*, **Pan X***, Chen H, Xu X, Zhang M. (2023). X00-responsive transcriptome reveals the role of the circular RNA133 in disease resistance by regulating expression of OsARAB in rice. Phytopathol Res 5, 22.
- Xia K*, **Pan X***, Zeng X, Zhang M. (2023). Rice miR168a-5p regulates seed length, nitrogen allocation and salt tolerance by targeting OsOFP3, OsNPF2.4, and OsAGO1a, respectively. J. Plant Physiol 208, 153905.
- Kelani AA, Bruch A, Riviaccio F, Visser C, Krüger T, Weaver D, **Pan X**, Schaeuble S, Panagiotou G, Kniemeyer O, Bromley MJ, Bowyer P, Barber AE, Brakhage AA, Blango MG. (2023). Disruption of the *A. fumigatus* RNA interference machinery alters the conidial transcriptome. RNA 29(7), 1033-1050.

CONFERENCES

- **2025 - Oral presentation:** German speaking Mycological Society (DMykG), Germany
- **2025 - Poster:** Gordon Research Seminar - RNA Editing, Italy
- **2024 - Oral presentation:** Jena RNA Club Symposium, Germany
- **2024 - Poster:** RNA Society Meeting, UK
- **2023 - Poster:** EMBL - The Non-coding Genome Symposium, Germany

AWARDS & SCHOLARSHIP

- DAAD Scholarship 2021 - 2025
- Jena School for Microbial Communication (JSMC) travel grant 2023

REFEREE

- Dr. Matthew Blango (matthew.blango@leibniz-hki.de)
- Dr. Olaf Kniemeyer (olaf.kniemeyer@leibniz-hki.de)